

WorkAtlas Frontend - Systems Architecture Documentation

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System Overview

WorkAtlas is an AI-powered job portal that connects job seekers with employers through intelligent matching. The frontend is built with Next.js 14, React, and a modern UI library for a responsive, and type-safe user interface.

Key Features

- **Dual User Roles:** Separate interfaces for Job Seekers and Employers
- **JWT Authentication:** Secure token-based authentication with persistent sessions
- **AI-Powered Matching:** Real-time job recommendations and candidate matching
- **Responsive Design:** Mobile-first design with Tailwind CSS
- **Real-time Updates:** Server-sent events for streaming recommendations

High-Level Architecture

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Architecture Layers

- 1. **Presentation Layer** (React Components)
 - Page components for routing
 - Reusable UI components
 - Lucide React icons for UI elements
- 2. **State Management Layer** (Zustand)
 - Authentication state (user, tokens, session)
 - Global application state
 - Persistent storage integration
- 3. **Service Layer** (API Services)
 - Auth service (login, register, token validation)
 - Seeker service (profile, jobs, applications)
 - Employer service (jobs, candidates, interviews)

Component	Responsibility
layout.tsx	Root layout, global styles, metadata
page.tsx	Marketing homepage, hero section
login/page.tsx	User authentication
register/page.tsx	User registration (seeker/employer)
seeker/dashboard/page.tsx	Seeker overview, stats, recent applications
seeker/jobs/page.tsx	Job search and filtering
seeker/recommendations/page.tsx	AI-powered job matches
seeker/applications/page.tsx	Application tracking
seeker/profile/page.tsx	Profile management, resume upload
employer/dashboard/page.tsx	Employer overview, job stats
employer/jobs/page.tsx	Job posting management
employer/jobs/new/page.tsx	Create new job posting
employer/jobs/[jobId]/candidates/page.tsx	AI candidate recommendations
employer/applications/page.tsx	Review applications
employer/profile/page.tsx	Company profile management

State Management

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Authentication Store Details

File: store/auth-store.ts

State Schema:

```
interface AuthState {
  user: User | null;           // Current user data
  isAuthenticated: boolean;    // Auth status
  isLoading: boolean;         // Loading state
  error: string | null;        // Error message

  // Actions
  setUser: (user: User | null) => void;
  setLoading: (loading: boolean) => void;
  setError: (error: string | null) => void;
  login: (token: string, user: User) => void;
  logout: () => void;
  checkAuth: () => Promise<void>;
}
```

Hook Usage:

- `useAuth()` : General authentication check
- `useSeekerAuth()` : Seeker-specific pages (auto-redirects if not seeker)
- `useEmployerAuth()` : Employer-specific pages (auto-redirects if not employer)

User Flow Diagrams

1. Job Seeker User Flow

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2. Employer User Flow

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3. Authentication Flow

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4. Job Application Flow

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5. AI Recommendation Flow

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6. Interview Scheduling Flow

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Technology Stack

Complete Frontend Tech Stack Overview

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Detailed Technology Breakdown

1. Core Framework & Runtime

Technology	Version	Purpose
Node.js	20 LTS	JavaScript runtime environment for server-side execution
Next.js	14.1.0	React framework with App Router, SSR, SSG, file-based routing, API routes

-  React Router integration (Next.js router)

2. Language & Type System

Technology	Version	Purpose
TypeScript	5.x	Statically typed superset of JavaScript for improved DX and type safety
@types/node	20.x	Type definitions for Node.js APIs
@types/react	18.x	Type definitions for React
@types/react-dom	18.x	Type definitions for React-DOM

TypeScript Configuration:

```
{
  "target": "ES2017",
  "lib": ["dom", "dom.iterable", "esnext"],
  "strict": true,
  "moduleResolution": "bundler",
  "jsx": "preserve",
  "paths": {
    "@/*": ["./*"] // Path aliasing for imports
  }
}
```

Key TypeScript Features:

-  Strict mode enabled
-  Path aliases (@/ for root imports)
-  Interface-based type definitions
-  Type inference
-  Generic types for API responses
-  Enum types for constants

3. Styling & Design System

Technology	Version	Purpose
Tailwind CSS	3.4.0	Utility-first CSS framework for rapid UI development
PostCSS	8.x	CSS transformation tool
Autoprefixer	10.x	Automatic vendor prefix addition

Tailwind Configuration:

- Custom color palette (primary blue shades)
- Custom theme extensions

```
}  
}
```

4. State Management & Data Fetching

Technology	Version	Purpose
Zustand	4.5.0	Lightweight state management library with minimal boilerplate
Axios	1.6.5	Promise-based HTTP client with interceptors and request/response transformation

Zustand Features:

-  Simple API with hooks
-  No providers/wrappers needed
-  Built-in TypeScript support
-  Middleware support
-  DevTools integration
-  Persistence (via localStorage integration)

Zustand Store Structure:

```
// Global stores  
- auth-store.ts // Authentication state
```

Axios Configuration:

-  Request interceptors (JWT token injection)
-  Response interceptors (error handling)
-  Base URL configuration
-  Timeout configuration
-  Custom headers
-  Content-Type handling
-  FormData support

5. UI Components & Icons

Technology	Version	Purpose
Lucide React	0.553.0	Beautiful, customizable icon library with 1000+ icons

Lucide Icons Used:

- Navigation: Briefcase , Menu , X , ArrowLeft , ArrowRight
- User Actions: User , Users , Logout , Login
- Content: FileText , Upload , Edit , Plus , Search
- Status: CheckCircle , XCircle , Clock , Activity , Eye

ESLint Configuration:

- Next.js recommended rules
- React hooks rules
- TypeScript-specific rules
- Accessibility rules (a11y)

Code Quality Features:

- ☒ Automatic code linting
- ☒ Next.js best practices enforcement
- ☒ React best practices
- ☒ Accessibility checks
- ☒ Consistent code style

7. Build Tools & Package Management

Technology	Version	Purpose
NPM	(with Node.js)	Package manager for JavaScript dependencies
Next.js Built-in Tools	14.1.0	Webpack, SWC compiler, bundling

Build Features:

- ☒ Fast Refresh (Hot Module Replacement)
- ☒ SWC-based compilation (faster than Babel)
- ☒ Automatic code splitting
- ☒ Tree shaking
- ☒ Minification
- ☒ Source maps for debugging

NPM Scripts:

```
{
  "dev": "next dev",           // Development server
  "build": "next build",       // Production build
  "start": "next start",       // Production server
  "lint": "next lint"          // Run ESLint
}
```

8. Containerization & Deployment

Technology	Version	Purpose
Docker	Latest	Container platform for consistent deployment
Node.js (Docker)	20-slim	Lightweight Node.js base image

Category	Technologies
Core	Node.js, Next.js, React, React-DOM
Language	TypeScript + Type Definitions
Styling	Tailwind CSS, PostCSS, Autoprefixer
State & HTTP	Zustand, Axios
UI/Icons	Lucide React
Dev Tools	ESLint, eslint-config-next
Infrastructure	Docker, NPM
Total Dependencies	6 production + 6 development

Dependency Tree Visualization

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Tech Stack Decision Rationale

Why Next.js 14?

- ☒ App Router provides better data fetching patterns
- ☒ Built-in optimizations (images, fonts, code splitting)
- ☒ Server-side rendering for SEO
- ☒ File-based routing simplifies structure
- ☒ Active development and strong community

Why Zustand over Redux?

- ☒ Minimal boilerplate (3x less code)
- ☒ No context providers needed
- ☒ Better TypeScript integration
- ☒ Smaller bundle size (3KB vs 30KB)
- ☒ Simpler learning curve

Why Tailwind CSS?

- ☒ Rapid prototyping
- ☒ Consistent design system
- ☒ Smaller CSS bundle (purged in production)
- ☒ No CSS naming conflicts
- ☒ Responsive utilities out of the box

Why Axios over Fetch?

- ☒ Interceptors for auth token injection

Package	Size (Minified)	Size (Gzipped)
Zustand	3 KB	1.2 KB
Axios	14 KB	5 KB
Lucide React	2-5 KB	1-2 KB
Tailwind CSS	Varies	~10 KB
Total (approx)	~90-95 KB	~42-45 KB

Performance Score:

- ⚡ Lighthouse Performance: 90+
- 📦 Small bundle size (<100KB gzipped)
- 🚀 Fast Time to Interactive (TTI)
- 🎨 Optimized CSS delivery

Environment Configuration

Environment Variables:

```
# Frontend Environment
NODE_ENV=development
NEXT_TELEMETRY_DISABLED=1
NEXT_PUBLIC_BACKEND_URL=http://localhost:8000 # or http://backend:8000 in Docker

# Production
NODE_ENV=production
NEXT_PUBLIC_BACKEND_URL=https://api.workatlas.com
```

Configuration Files:

- next.config.js - Next.js configuration
- tsconfig.json - TypeScript compiler options
- tailwind.config.ts - Tailwind customization
- postcss.config.js - PostCSS plugins
- .eslintrc.json - ESLint rules
- package.json - Dependencies and scripts
- Dockerfile - Container configuration

Backend Integration Stack

Technology	Purpose	
FastAPI	Python backend framework	F
MongoDB	NoSQL database for user/job data	E
ChromaDB	Vector database for embeddings	E

API Service Structure

Authentication Service (services/auth-service.ts)

Method	Endpoint	
login()	POST /auth/login	Authenticate user
registerSeeker()	POST /auth/register/seeker	Create new job seeker
registerEmployer()	POST /auth/register/employer	Create new employer
validateToken()	GET /auth/me	Validate JWT token

Seeker Service (services/seeker-service.ts)

Method	Endpoint	
getProfile()	GET /seekers/me	Get current user profile
updateProfile()	PUT /seekers/me	Update current user profile
uploadResume()	POST /seekers/resume	Upload resume
applyForJob()	POST /seekers/applications	Submit job application
getApplications()	GET /seekers/applications	Get all applications
withdrawApplication()	DELETE /seekers/applications/{id}	Withdraw application
searchJobs()	GET /seekers/jobs	Search for jobs
getJobDetails()	GET /seekers/jobs/{id}	Get specific job details
getRecommendations()	GET /seekers/recommendations	Get AI-generated recommendations
getAutoApplySettings()	GET /seekers/auto-apply/settings	Get auto-apply settings
updateAutoApplySettings()	POST /seekers/auto-apply/settings	Update auto-apply settings

Employer Service (services/employer-service.ts)

Method	Endpoint	
getProfile()	GET /employers/me	Get current employer profile
updateProfile()	PUT /employers/me	Update current employer profile
createJob()	POST /employers/jobs	Create new job listing

```

// API Client Structure (Inferred from usage)
import axios from 'axios';

// Determine API URL based on environment
const getApiUrl = () => {
  if (typeof window !== 'undefined') {
    // Browser context - use Localhost
    return 'http://localhost:8000';
  }
  // SSR context - use Docker hostname if available
  return process.env.NEXT_PUBLIC_BACKEND_URL || 'http://localhost:8000';
};

const apiClient = axios.create({
  baseURL: `${getApiUrl()}/api/v1`,
  headers: {
    'Content-Type': 'application/json',
  },
  timeout: 30000, // 30 seconds default
});

// Request interceptor - Add JWT token to requests
apiClient.interceptors.request.use(
  (config) => {
    if (typeof window !== 'undefined') {
      const token = localStorage.getItem('access_token');
      if (token) {
        config.headers.Authorization = `Bearer ${token}`;
      }
    }
    return config;
  },
  (error) => Promise.reject(error)
);

// Response interceptor - Handle errors
apiClient.interceptors.response.use(
  (response) => response,
  (error) => {
    if (error.response?.status === 401) {
      // Token expired - redirect to login
      if (typeof window !== 'undefined') {
        localStorage.removeItem('access_token');
        localStorage.removeItem('user_role');
        localStorage.removeItem('user_id');
        window.location.href = '/login';
      }
    }
    return Promise.reject(error);
  }
);

export default apiClient;

```

Request/Response Flow

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Key Features Implementation

1. JWT Authentication & Session Management

- Token Storage: `localStorage( access_token , user_role , user_id )`

- **Adaptive Layouts:** Grid systems that adjust to screen size
- **Touch-Friendly:** Large interactive elements on mobile
- **Progressive Enhancement:** Works on all device sizes

5. Real-time Features

- **Application Status:** Real-time updates on application progress
 - **Live Stats:** Dashboard metrics update dynamically
 - **Streaming Recommendations:** Progressive loading of AI results
 - **Email Notifications:** n8n integration for interview scheduling
-

Security Considerations

1. **JWT Token Security**
 - Tokens stored in localStorage (accessible only to same origin)
 - Short-lived tokens (configurable expiry)
 - Automatic cleanup on logout
 - Token validation on every protected route
 2. **Route Protection**
 - All user pages require authentication
 - Role-based access control enforced
 - Automatic redirects for unauthorized access
 3. **API Security**
 - Bearer token authentication
 - CORS configuration for allowed origins
 - Request/response validation
 - Error handling without exposing sensitive data
 4. **Data Privacy**
 - Sensitive data transmitted over HTTPS (production)
 - User passwords never stored in frontend
 - Profile data accessible only to authenticated users
-

Performance Optimizations

1. **Code Splitting**
 - Next.js automatic code splitting per route
 - Lazy loading of non-critical components
 - Dynamic imports for heavy features
2. **Caching Strategy**
 - sessionStorage for recommendation caching
 - localStorage for auth persistence
 - API response caching where appropriate
3. **Asset Optimization**

```
├── seeker/                # Seeker pages
│   ├── dashboard/
│   ├── jobs/
│   ├── recommendations/
│   ├── applications/
│   └── profile/
├── employer/             # Employer pages
│   ├── dashboard/
│   ├── jobs/
│   ├── applications/
│   └── profile/
├── components/           # Reusable components (future)
├── hooks/                # Custom React hooks
│   └── useAuth.ts
├── lib/                  # Utilities and configs
│   └── api.ts             # API client (needs creation)
├── services/             # API service layer
│   ├── auth-service.ts
│   ├── seeker-service.ts
│   └── employer-service.ts
├── store/                # State management
│   └── auth-store.ts
├── types/                # TypeScript types
│   └── index.ts
├── styles/               # Global styles
│   └── globals.css
```

Naming Conventions

- **Components:** PascalCase (e.g., `SeekerDashboard`)
- **Files:** kebab-case for folders, PascalCase for components
- **Functions:** camelCase (e.g., `getProfile`)
- **Constants:** UPPER_SNAKE_CASE (e.g., `API_URL`)
- **Types/Interfaces:** PascalCase (e.g., `User` , `AuthState`)

Code Style

- Use TypeScript for all new code
- Functional components with hooks
- Explicit return types for functions
- Comprehensive JSDoc comments for services
- Tailwind for styling (no CSS modules)

Future Enhancements

1. **Component Library**
 - Extract reusable UI components
 - Create shared component library
 - Implement Storybook for component documentation
2. **Advanced Features**
 - Real-time notifications (WebSockets)
 - Chat system between seekers and employers
 - Advanced filtering and search
 - Saved searches and job alerts

Conclusion

The WorkAtlas frontend is a modern, type-safe, and performant React application built with Next.js 14. It implements robust authentication, role-based access control, and maintains a clean architecture and excellent developer experience.

The modular service layer, centralized state management, and custom hooks provide a solid foundation for future enhancements and scalability.

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